Exp 11;Write the python program for Map Coloring to implement CSP

```
colors = ["Red", "Green", "Blue"]

graph = {
    "A": ["B", "C"],
    "B": ["A", "C", "D"],
    "C": ["A", "B", "D"],
    "D": ["B", "C"]
}

assigned = {}

def solve(node):
    if node == len(graph):
        return True

    region = list(graph.keys())[node]

    for color in colors:
        if all(assigned.get(n) != color for n in graph[region]):
            assigned[region] = color

            if solve(node + 1):
                return True

            del assigned[region]

    return False

if solve(0):
    print("Map Coloring Solution:")
    print(assigned)
else:
    print("No solution exists")
```

Output

```
Map Coloring Solution:
{'A': 'Red', 'B': 'Green', 'C': 'Blue', 'D': 'Red'}
```